5 **(a)** State Faraday's law of electromagnetic induction.

.....

.....

..... [1]

(b) Fig. 5.1 shows a coil of 400 turns and cross-sectional area of 4.2×10^{-4} m^2 placed in the middle of a long solenoid. The coil is connected to a sensitive ammeter. The solenoid is connected to a variable resistor and a sinusoidally-alternating voltage supply.

The variation with time t of the magnetic flux density B in the solenoid is shown in Fig. 5.2.

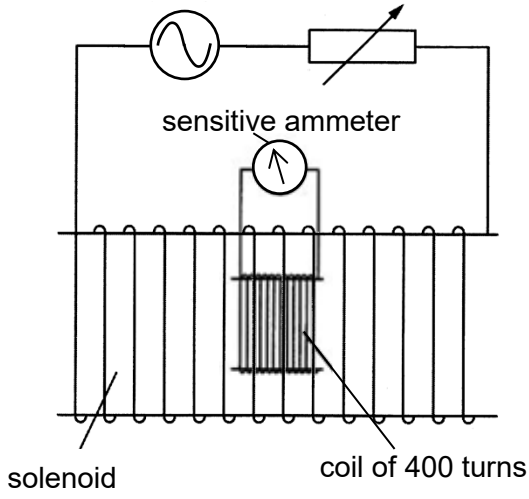

Fig. 5.1

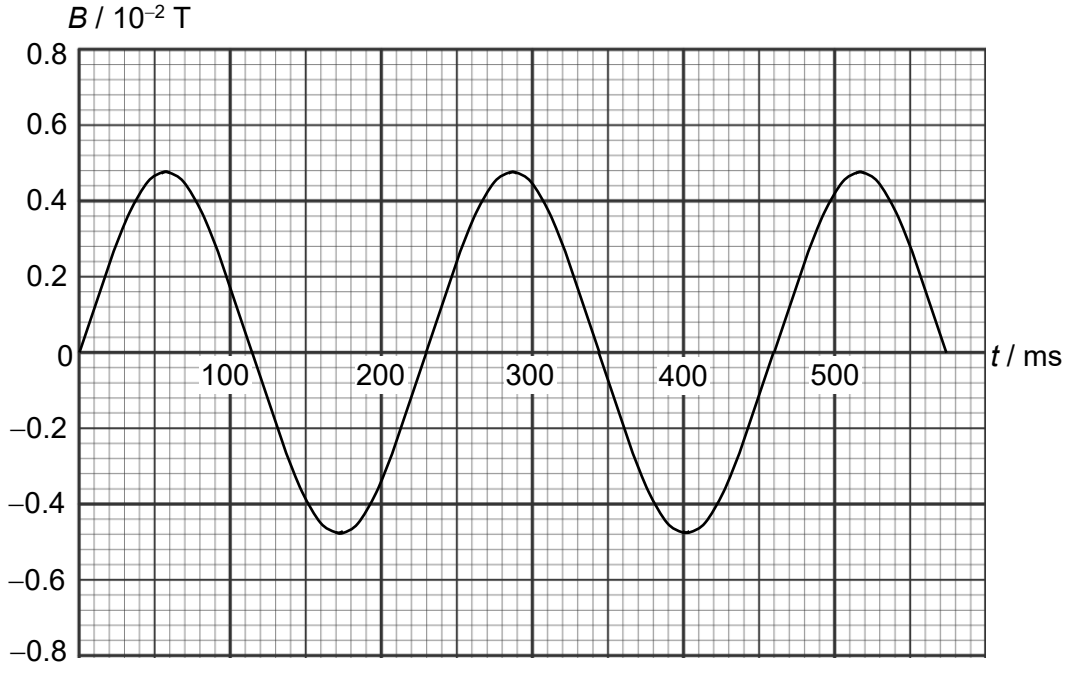

Fig. 5.2

(i) Explain, with reference to the magnetic field in the solenoid, why the sensitive ammeter deflects in opposite directions.

.....

.....

.....

.....

..... [2]

(ii) Using Fig. 5.2,

1. state a value of t when the magnitude of the induced e.m.f. is maximum,

t = ... s [1]

2. hence determine the maximum induced current in the coil given that the resistance of the coil is 5.0 Ω,

maximum induced current = ... A [3]

3. calculate the mean power dissipated by the coil.

mean power dissipated = ... W [2]

(iii) Sketch on Fig. 5.3, a graph to show the variation with time t of the induced current i in the coil from t = 0 ms to t = 460 ms. Include appropriate scale markings on the vertical axis.

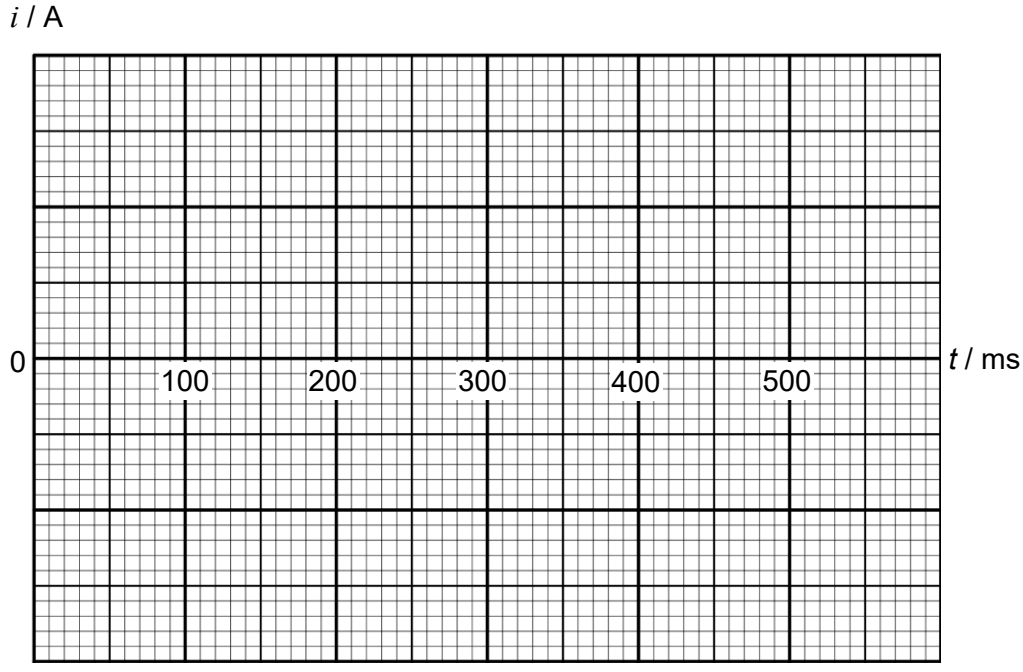

Fig. 5.3

[2]

(iv) Sketch on Fig. 5.4, a graph to show the variation with time t of the power P dissipated in the coil from t = 0 ms to t = 460 ms. Include appropriate scale markings on the vertical axis.

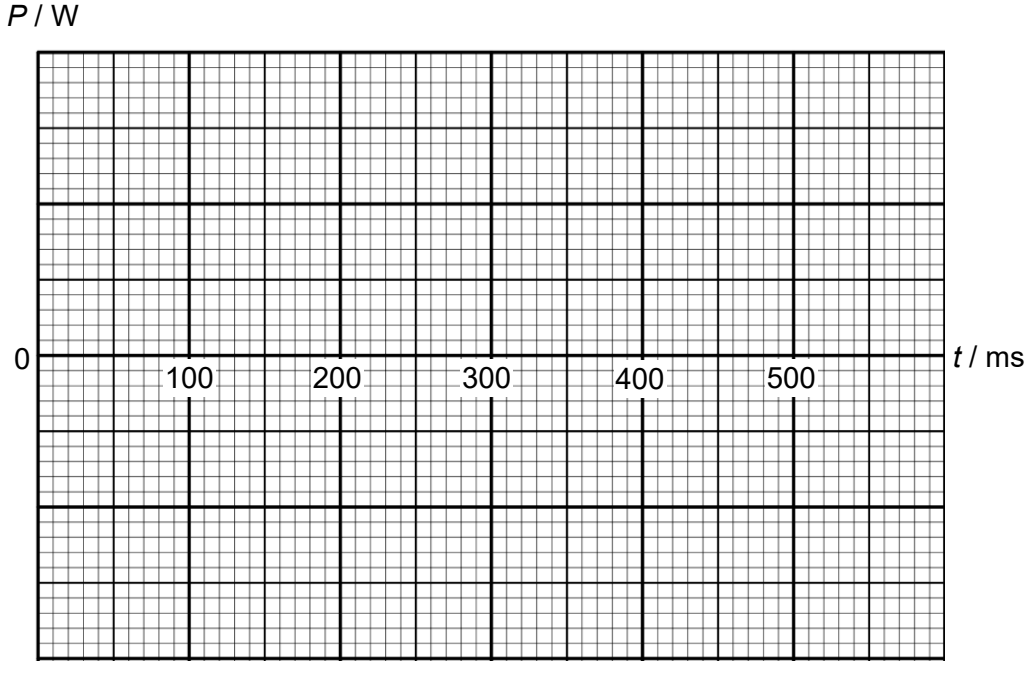

Fig. 5.4

[2]

End of Paper 3 Section A